SIMATS ENGINEERING

ASSESSMENT TOOL 2-Pseudocode Writing Task

COURSE NAME: R PROGRAMMING **COURSE CODE: ITA04**

COURSE OUTCOME COVERED

CO1: Apply R programming fundamentals including data structures, vectors, matrices, arrays, and class types to perform mathematical operations and manage data effectively. Identify and implement appropriate control structures to solve a particular programming problem
(Bloom Level -BL3- BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 Questions **Total Marks: 100**

Code of Conduct:

I A.Midhuna (192521302) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

A. Midhuna.

Signature of the student:

Instructions to students: Write structured pseudocode (not R syntax) for each scenario. Clearly show inputs, data structures used, control structures (decision, loop), processing steps, and outputs. Marks: 10 per question.

Q1. Campus Canteen Billing (BL3: Vectors)

The campus canteen stores item prices in a numeric vector price and the quantity ordered in a vector qty for one customer. Write pseudocode to compute the bill for each item, the total bill, and apply a 10% discount if the total exceeds ₹500. Display the final payable amount.

Expected concepts: vector arithmetic, element-wise multiplication, sum, if-else.

Q2. Student Marks Analyser (BL3: Vectors, Control Structures)

A teacher stores the marks of 40 students in a vector. Write pseudocode to find (a) the class average, (b) the highest and lowest mark, and (c) the count of students who scored below 40 (fail). Print "Class Performance: Good" if more than 80% have passed, otherwise "Needs Improvement".

Expected concepts: vector functions, logical indexing, counting with condition, if-else.

Q3. Hostel Room Allocation Matrix (BL3: Matrices)

A hostel has 4 floors and 5 rooms per floor. A 4×5 matrix stores 1 if a room is occupied and 0 if vacant. Write pseudocode to display (a) the number of vacant rooms on each floor, (b) the total vacant rooms in the hostel, and (c) the floor with the maximum vacancies.

Expected concepts: matrix creation, row-wise sums, nested loops or apply-style logic, finding maximum.

Q4. Monthly Electricity Bill Calculator (BL3: Control Structures)

An electricity board charges ₹3/unit for the first 100 units, ₹5/unit for the next 100 units, and ₹8/unit beyond 200 units. Given the units consumed by a household, write pseudocode to compute the total bill. Add a fixed charge of ₹50 and print the bill summary.

Expected concepts: nested if-else / if-else-if ladder, slab-based calculation.

Q5. Sales Report Using Matrix Operations (BL4: Matrices, Analysis)

A shop records the sales of 3 products over 4 weeks in a 3×4 matrix. Write pseudocode to (a) compute the total sales per product, (b) compute the weekly total sales, (c) identify the best-performing product and the best week, and (d) transpose the matrix so weeks appear as rows.

Expected concepts: matrix row/column sums, which.max-type logic, transpose, analysis of results.

Assessment Tool - 2

Pseudocode writing Task

A. Midhuna
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1. START

```
Input price[]
Input qty[]
bill = price * qty
total = sum(bill)
IF total > 500 THEN
  discount = total * 0.10
ELSE
   discount = 0
END IF
final_amount = total - discount
Display "Bill for each item:", bill
Display "Total Bill:", total
Display "Discount:", discount
Display "Final payable Amount:", final_amount
END.
```

2. Student marks Analyser

```
START
Input marks[40]
average ← sum(marks)/40
highest ← MAX(marks)
```

```
lowest ← MIN(marks)
fail_count ← COUNT(marks <40)
passed_count ← 40 - fail_count
pass_percentage ← (passed_count /40) x 100
IF pass_percentage >80 THEN
   DISPLAY "class performance : Good"
ELSE
   DISPLAY "class performance : needs Improvement"
END IF
DISPLAY "class Average:", average
DISPLAY "Highest mark:", highest
DISPLAY "Lowest mark:", lowest
DISPLAY "number of students failed:", fail_count
DISPLAY "pass percentage:", pass_percentage
END
```

3. Hostel Room Allocation matrix

```
START
Input room[4][5]
FOR i←1TO4
  vacant[i]←0
 FOR j←1TO5
```

```
        room[i][j] = 0 THEN
            vacant[i] ← vacant[i] + 1
    END IF
END FOR
DISPLAY "Vacant rooms on Floor ", i, ": ", vacant[i]
END FOR
total_vacant ← SUM(vacant)
max_vacant ← MAX(vacant)
floor_max ← POSITION(max_vacant IN vacant)
DISPLAY "Total vacant Rooms: ", total_vacant
DISPLAY "Floor with maximum vacancies: ",
                                floor_max
DISPLAY "maximum vacancies: ", max_vacant
END
```

4. Monthly Electricity Bill calculator

```
START
Input units
IF units <= 100 THEN
  energy_charge ← units x 3
ELSE IF units <= 200 THEN
  energy_charge ← (100 x 3) + ((units - 100) x 5)
ELSE
```

```
energy_charge ← (100 x 3) + (100 x 5) + ((units - 200) x
END IF
fixed_charge ← 50
total_bill ← energy_charge + fixed_charge
DISPLAY "Electricity Bill Summary"
DISPLAY "Units consumed:", units
DISPLAY "Energy charge: ₹", energy_charge
DISPLAY "Fixed charge: ₹", fixed_charge
DISPLAY "Total Bill: ₹", total_bill
END
```

5. Sales Report using matrix operations

```
START
Input Sales[3][4]
    3 products x 4 weeks
(a) Total Sales per product
FOR i ← 1 TO 3
Product_total[i] ← SUM(sales[i][1...4])
END FOR
DISPLAY "Total Sales per product :", product_total
(b) Weekly total sales
FOR j ← 1 TO 4
  week_total[j] ← 0
```

```
FOR i ← 1 TO 3
    week_total[j] ← week_total[j] + sales[i][j]
END FOR
DISPLAY "weekly Total sales:", week_total
```

(c) Best-performing product and best week

```
best_product ← POSITION(MAX(product_total) IN
                                    product_total)
best_week ← POSITION(MAX(week_total) IN week_total)
DISPLAY "Best performing product:", best_product
DISPLAY "Best week:", best_week
```

(d) Transpose the matrix

```
transposed_Sales ← TRANSPOSE(Sales)
DISPLAY "Transposed Sales Matrix:"
DISPLAY transposed_Sales
END
```